

Name: _____

Class: _____

2017

Term 4 Assessment (HSC)

Mathematics Extension 2

General Instructions

- Working time – 90 minutes + 5minutes reading
- Board approved calculators may be used
- Write using black or blue pen
- All necessary working should be shown in all questions
- Write your student number at the top of every page of your answer sheets.

4 questions – 15 marks each

Total marks – 60

Attempt all questions

Start each question on a new sheet of paper.

QUESTION 1 (Start a new page)**15 Marks**

- a) Let $z = 1 + 2i$ and $w = 2 - i$. Find in the form $x + iy$
- i) $z\bar{w}$ **2**
- ii) $\frac{1}{w^2}$ **2**
- b) Let θ be a real number and consider $(\cos\theta + i\sin\theta)^3$.
Prove $\cos 3\theta = \cos^3\theta - 3\cos\theta \sin^2\theta$ **2**
- c) Sketch on the Argand diagram the region which satisfies these expressions
 $2 \leq |z| \leq 3$ and $\frac{\pi}{4} < \arg(z - i) < \frac{3\pi}{4}$ **3**
- d) $z = x + iy$ is such that $\frac{z-i}{z+1}$ is purely imaginary. Find the equation of the locus of the point P representing z . Describe this locus geometrically. **3**
- e) Given that i is a zero of $P(x) = x^4 - x^3 + 3x^2 - x + 2$, find all complex solutions to $x^4 - x^3 + 3x^2 - x + 2 = 0$. **3**

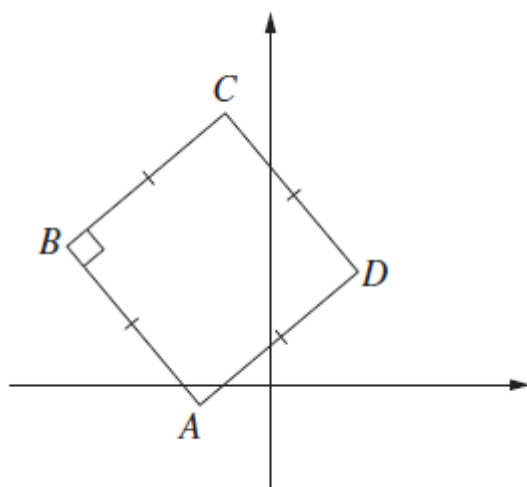
QUESTION 2 (Start a new page)**15 Marks**

- a) i) Find the square roots of $6i$. **2**
- ii) Hence or otherwise solve $z^2 - (1 - i)z - 2i = 0$ **2**
- b) The polynomial equation $x^3 - 2x^2 + 4x - 5 = 0$ has roots α, β and γ .
- i. Find the value of $\frac{1}{\alpha^3\beta^3\gamma^3}$ **1**
- ii. Which polynomial equation has roots α^2, β^2 and γ^2 ? **2**
- iii. Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. **2**
- c) Let $z_1 = 4 + i$ and $z_2 = 1 + i$.
- i) Sketch the locus of z on the Argand plane such that $\arg\left(\frac{z-z_2}{z-z_1}\right) = \frac{\pi}{2}$ **2**
- ii) Explain why $\arg\left(z - \frac{5}{2} - i\right) = 2\arg(z - 1 - i)$. **2**
- d) The polynomial $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$ has a root of multiplicity 3. Find all the roots of $P(x)$ and hence factorise $P(x)$. **2**

QUESTION 3 (Start a new page)

15 Marks

- a) If w is a complex cube root of unity, simplify $(1 + w - w^2)^7$ in terms of w^2 . **1**
- b) Find, in modulus-argument form, the three cube roots of -8 **2**
- c) The points A, B, C and D on the Argand diagram below represent the complex numbers a, b, c and d respectively.



The points form a square as shown on the diagram above. By using vectors, or otherwise, show that $d = a(1 + i) - ib$. **2**

- d) i) On an Argand diagram, shade in the region determined by the inequalities **3**

$$1 \leq \operatorname{Im} z \leq 2 \text{ and } \frac{5\pi}{12} \leq \arg z \leq \frac{11\pi}{12}$$

- ii) Let z_0 be the complex number of maximum modulus satisfying the inequalities of d(i). Express z_0 in the form of $a + ib$ in exact form. **2**

- e) Let $z = \cos\theta + i\sin\theta$.

i) Show that $\sin n\theta = \frac{z^n - z^{-n}}{2i}$ and $\cos n\theta = \frac{z^n + z^{-n}}{2}$ **2**

- ii) Hence, or otherwise, prove the identity $32\sin^4\theta\cos^2\theta = \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2$ **3**

QUESTION 4 (Start a new page)**15 Marks**

- a) If $|z| = |w|$, prove that $\frac{z+w}{z-w}$ is purely imaginary. By drawing a suitable diagram, give a geometrical interpretation of the result. **3**
- b) i) Write $\sqrt{6} - \sqrt{2}i$ in modulus-argument form. **2**
- ii) Hence, expand and simplify $\left(\frac{\sqrt{6}-\sqrt{2}i}{2-2i}\right)^{2000}$. Give your answer in the form of $x + iy$ **2**
- c) Let α be the complex root of the polynomial $z^7 = 1$ with the smallest possible argument. Let $\theta = \alpha + \alpha^2 + \alpha^4$ and $\delta = \alpha^3 + \alpha^5 + \alpha^6$
- i) Explain why $\alpha^7 = 1$ and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$ **1**
- ii) Show that $\theta + \delta = -1$ and $\theta\delta = 2$ and hence write a quadratic equation whose roots are θ and δ **3**
- iii) Write down α in mod-arg form, and show that $\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$ and $\sin \frac{4\pi}{7} + \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$ **4**

******END OF EXAMINATION******

Y12 Task 1: 2017		MATHEMATICS Extension 2: Question 1		JRAHS
Suggested Solutions		Marks	Marker's Comments	
a)	$(i) \quad z \bar{w} = (1+2i)(2+i)$ $= 2+5i-2$ $= 0+5i$	1	correct conjugate for w	
		1	correct product	
	$(ii) \quad \frac{1}{w^2} = \frac{1}{(2-i)^2} = \frac{1}{4-4i-1}$ $= \frac{1}{3-4i} \times \frac{3+4i}{3+4i}$ $= \frac{3+4i}{9+16}$ $= \frac{3}{25} + \frac{4}{25}i$	1	For squaring and realising the denominator	
		1	For correct simplified expression	
b)	$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta \quad \text{--- (1) by De Moivre's Theorem}$	1	For correct reason	
	But $(\cos \theta + i \sin \theta)^3$ $= \cos^3 \theta + 2 \cos^2 \theta i \sin \theta + 2 \cos \theta i^2 \sin^2 \theta + i^3 \sin^3 \theta$ $= \cos^3 \theta - 2 \cos \theta \sin^2 \theta + i(2 \cos^2 \theta \sin \theta - \sin^3 \theta) \quad \text{--- (2)}$		Using Binomial Theorem or just cubing of a binomial	
	Equating real parts of (1) and (2), it can be seen that $\cos 3\theta = \cos^3 \theta - 2 \cos \theta \sin^2 \theta$	1	Correctly stating "Equate Reals"	

Y12 Task 1 : 2017

MATHEMATICS Extension 2: Question 1

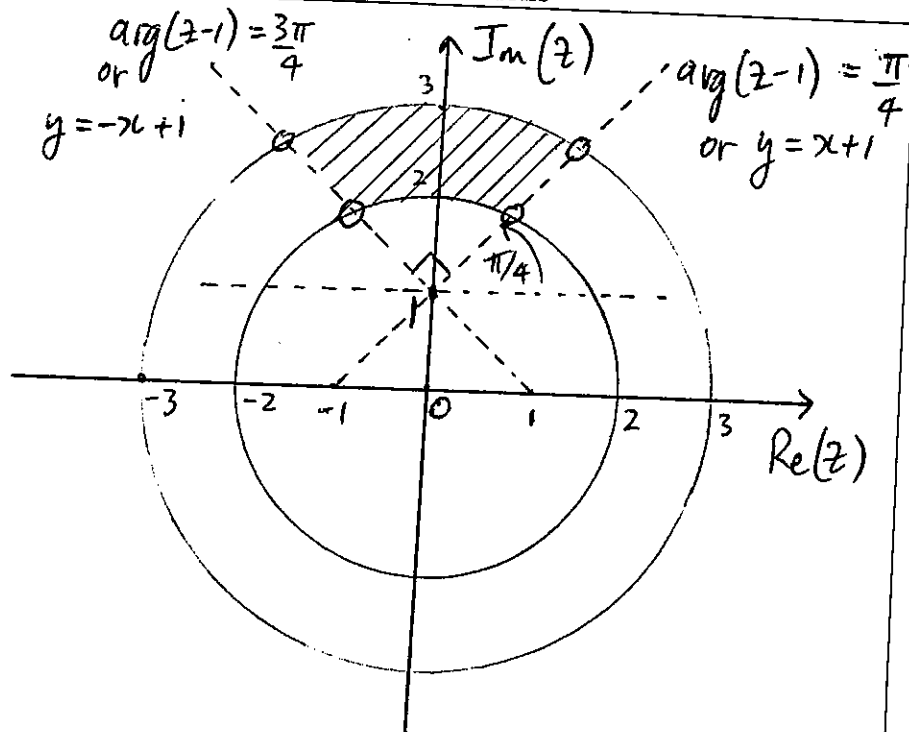
JRAHS

Suggested Solutions

Marks

Marker's Comments

c)



1

For correct shaded region

1

For correct arguments

1

For correct boundaries

If 2 or more errors, then max 1

If no arg marked on diagram, max 2

Y12: Task 1: 2017

MATHEMATICS Extension 2: Question 1

JRAHS

Suggested Solutions

Marks

Marker's Comments

d)

Method 1:

$$\begin{aligned} \frac{z-i}{z+1} &= \frac{x+i(y-1)}{x+1+iy} \times \frac{x+1-iy}{x+1-iy} \\ &= \frac{x(x+1)+y(y-1)+i(x+1)(y-1)-ixy}{(x+1)^2+y^2} \end{aligned}$$

For $\frac{z-i}{z+1}$ to be purely imaginary,

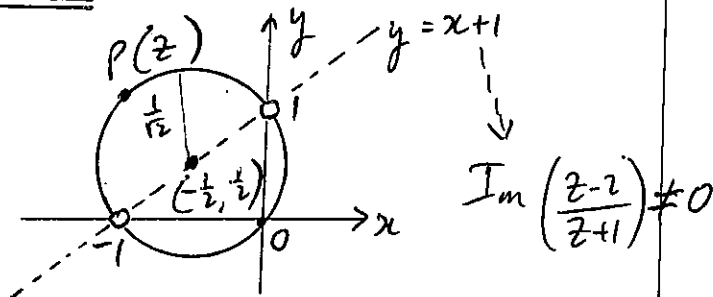
$$\operatorname{Re}\left(\frac{z-i}{z+1}\right) = 0$$

$$\Rightarrow \frac{x(x+1)+y(y-1)}{(x+1)^2+y^2} = 0$$

$$\Rightarrow x^2+x+y^2-y=0$$

$$\text{i.e. } \left(x+\frac{1}{2}\right)^2 + \left(y-\frac{1}{2}\right)^2 = \frac{1}{2}$$

$\therefore$ Focus is a circle with centre $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and radius $\frac{1}{\sqrt{2}}$, with $(-1,0)$ and $(1,0)$ excluded.

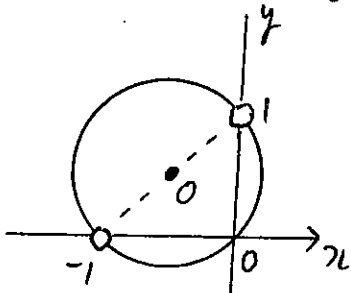


For realising denominator

Equation of locus

Geometrical description

If $\frac{z-i}{z+1}$, then max 1

Y12 : Task 1 : 2017 MATHEMATICS Extension 2: Question 1		JRAHS
Suggested Solutions	Marks	Marker's Comments
d) <u>Method 2</u> For $\frac{z-i}{z+1}$ to be purely imaginary, then $\arg\left(\frac{z-i}{z+1}\right) = \arg(z-i) - \arg(z+1) = \pm\frac{\pi}{2}$  $\therefore$ Coordinates of centre $(-\frac{1}{2}, \frac{1}{2})$, with radius $\frac{1}{\sqrt{2}}$ and hence a circle, excluding $(-1, 0)$ and $(0, 1)$, with equation $(x+\frac{1}{2})^2 + (y-\frac{1}{2})^2 = \frac{1}{2}$ <u>Method 3 :</u> Let $\frac{z-i}{z+1} = ki \dots$ purely imaginary $\Rightarrow k = -\frac{x}{y}$ and $k = \frac{y-1}{x+1}$ $\hookrightarrow x^2 + x + y^2 - 1 = 0$ $\hookrightarrow (x+\frac{1}{2})^2 + (y-\frac{1}{2})^2 = \frac{1}{2}$ etc, etc.	1 1 1 1 1 1 1	For correct argument Equation of locus Geometrical description <u>Note :</u> No marks were awarded for the restrictions description

Y12 : Task 1 : 2017 MATHEMATICS Extension 2: Question 1		TRAHS	
Suggested Solutions		Marks	Marker's Comments
e) Since $P(x)$ has real coefficients, by the conjugate root theorem, complex roots appear in conjugate pairs i.e. Since $P(i)=0$, then $P(-i)=0$ $\Rightarrow \pm i$ are roots of $P(x)$ i.e. $(x+i)(x-i) = x^2+1$ are factors of $P(x)$. Hence $P(x) = (x^2+1)Q(x)$ $Q(x)$ obtained by long division or by inspection $\therefore P(x) = (x^2+1)(x^2-x+2)$ $= (x+i)(x-i)(x-\frac{1}{2})^2 + \frac{7}{4}$ $= (x+i)(x-i)(x-\frac{1}{2} + \frac{\sqrt{7}i}{2})(x-\frac{1}{2} - \frac{\sqrt{7}i}{2})$ $\therefore$ The roots are $\pm i, \quad \frac{1}{2} \pm \frac{\sqrt{7}i}{2}$		1	For $-i$ is a root and that x^2+1 is a factor
		1	For the other factor of $P(x)$
		1	For stating all the roots

Y12	Task 1 : 2017 MATHEMATICS Extension 2: Question 1	JRAHS
	Suggested Solutions	Marks
e) <u>Method 2</u> : $P(i) = 0 \dots$ given $P(-i) = 0 \dots$ $P(x)$ has real coefficients and hence roots come in conjugate pairs. Let $\alpha, \bar{\alpha}$ be the other roots $\sum \alpha = 1 \Rightarrow \alpha + \bar{\alpha} = 1$ $\Rightarrow 2x = 1$ if $\alpha = x + iy$ $x = \frac{1}{2}$ $\sum \alpha \beta \gamma \delta = 2 \Rightarrow \alpha \bar{\alpha} = 2$ $\Rightarrow x^2 + y^2 = 2$ $\therefore y = \pm \frac{\sqrt{7}}{2}$ $\therefore$ The roots are $\pm i, \frac{1}{2} \pm \frac{\sqrt{7}}{2} i$	1 1 1	Identifying $-i$ is also a root For obtaining 2 of the roots For stating all the roots

Suggested Solutions

Marks

Marker's Comments

a) i) Let $z = x + iy$

$$z^2 = (x^2 - y^2) + 2xyi$$

$$z^2 = 6i \Rightarrow x^2 - y^2 = 0 \quad \text{Eqn 1}$$

$$2xy = 6$$

$$xy = 3$$

$$y = \frac{3}{x} \quad \text{Eqn 2}$$

Equating
Re(z) and Im(z)

①

Sub Eqn 2 into Eqn 1

$$x^2 - \frac{9}{x^2} = 0$$

$$x^4 - 9 = 0$$

$$(x^2 + 3)(x^2 - 3) = 0$$

$$x = \pm\sqrt{3} \quad (x \in \mathbb{R})$$

when $x = \sqrt{3}$, $y = \sqrt{3}$,

$x = -\sqrt{3}$, $y = -\sqrt{3}$

Final
Answer

$$\therefore \sqrt{6}i = \sqrt{3} + \sqrt{3}i \quad \text{or} \quad -\sqrt{3} - \sqrt{3}i$$

①

ii) $z = \frac{(1-i) \pm \sqrt{(1-i)^2 - 4(1)(2i)}}{2}$

$$= \frac{(1-i) \pm \sqrt{-2i + 8i}}{2}$$

$$= \frac{(1-i) \pm \sqrt{6}i}{2} \quad \text{Getting to this step}$$

①

$$= \frac{(1-i) \pm (\sqrt{3} + \sqrt{3}i)}{2} = \frac{(1+\sqrt{3})}{2} + \frac{(\sqrt{3}-1)}{2}i \quad \text{or} \quad \frac{1-\sqrt{3}}{2} - \frac{(1+\sqrt{3})}{2}i$$

①

Suggested Solutions

Marks

Marker's Comments

b) i $\frac{1}{\alpha^3 \beta^3 \gamma^3} = \frac{1}{(\alpha \beta \gamma)^3} = \frac{1}{5^3} = \frac{1}{125}$

(1)

ii Method 1

Polynomial will have equation

$(\sqrt{x})^3 - 2(\sqrt{x})^2 + 4\sqrt{x} - 5 = 0$ Sub $\sqrt{x}$ into eqn (1)

$x\sqrt{x} + 4\sqrt{x} = 5 + 2x$

$x^3 + 8x^2 + 16x = 25 + 20x + 4x^2$

$x^3 + 4x^2 - 4x - 25 = 0$ answer

(1)

Method 2

$\alpha^2 \beta^2 \gamma^2 = (\alpha \beta \gamma)^2 = 5^2 = 25$

$(\alpha^2 + \beta^2 + \gamma^2) = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= (2)^2 - 2(4)$

$= -4$

Need Both values

(1)

$\alpha^2 \beta^2 + \alpha^2 \gamma^2 + \beta^2 \gamma^2 = (\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2(\alpha^2 \beta \gamma + \alpha \beta^2 \gamma + \alpha \beta \gamma^2)$

$= (\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$= 4^2 - 2(5)(2)$

$= 16 - 20$

$= -4$

$\therefore$ Polynomial is given by $x^3 + 4x^2 - 4x - 25 = 0$

(1)

Suggested Solutions

Marks

Marker's Comments

iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

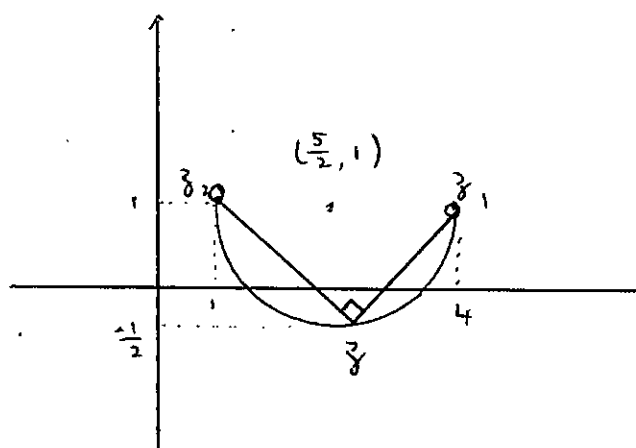
$= \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2}$

(1)

$= \frac{-4}{25}$ From equation found in part ii)

(1)

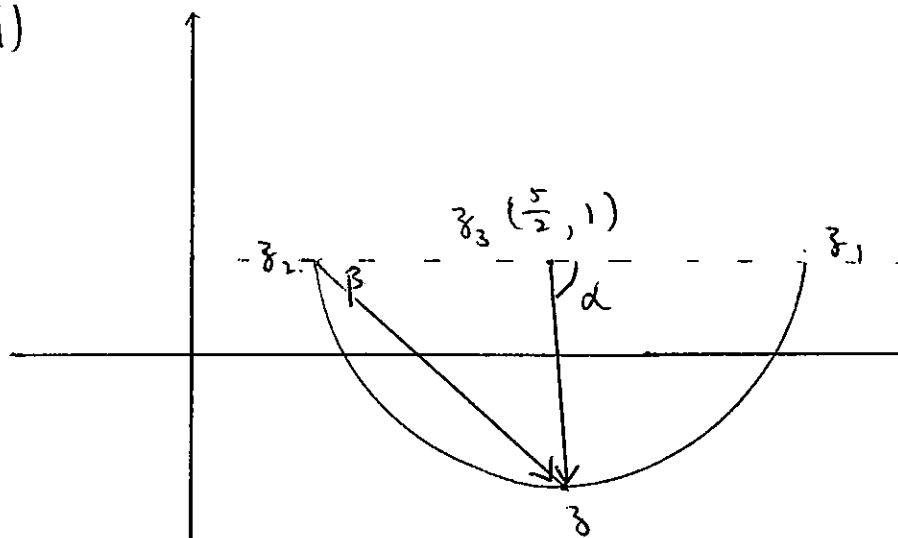
c) i



- ① — Clearly demonstrating a semi-circle By:
- Marking a centre or
 - Marking the angle subtended at z

- ① — Correctly getting the bottom half
- +
open dots at z_1 and z_2
- +
correct scale

ii)



$$\begin{aligned} \text{Arg}(z - \frac{5}{2} - i) &= \text{Arg}(z - (\frac{5}{2} + i)) \\ &= \alpha \text{ (see diagram)} \end{aligned}$$

$$\begin{aligned} \text{Arg}(z - 1 - i) &= \text{Arg}(z - (1 + i)) \\ &= \beta \text{ (see diagram)} \end{aligned}$$

Method 1

$\alpha = 2\beta$ (angle at the centre is twice the angle at the circumference standing on the same arc).

$$\therefore \text{Arg}(z - \frac{5}{2} - i) = 2\text{Arg}(z - 1 - i)$$

Both are needed

Method 2

$z_3 z_2 = z_3 z$ (radii of a circle)

$\therefore \angle z_3 z z_2 = \beta$ (equal angles are opposite to equal sides in $\Delta z_2 z_3 z$)

$\therefore \alpha = 2\beta$ (exterior angle of $\Delta z_2 z_3 z$)

①

Correctly demonstrating what $\text{Arg}(z - \frac{5}{2} - i)$ and $\text{Arg}(z - 1 - i)$ is.

①

①

Suggested Solutions

Marks

Marker's Comments

$$d) P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$$

$$P'(x) = 4x^3 - 9x^2 - 12x + 28$$

$$P''(x) = 12x^2 - 18x - 12$$

$$= 6(2x^2 - 3x - 2)$$

$$= 6(x-2)(2x+1)$$

$$\therefore P''(x) = 0 \Rightarrow x = 2 \text{ or } -\frac{1}{2}$$

①

$$P(2) = (2)^4 - 3(2)^3 - 6(2)^2 + 28(2) - 24$$

$$= 16 - 24 - 24 + 56 - 24$$

$$= 0$$

$\therefore x=2$ is a root with multiplicity 3.

$\therefore$ Let roots of $P(x)$ be 2, 2, 2 and α

$$2+2+2+\alpha = 3 \text{ (sum of roots)}$$

$$6+\alpha = 3$$

$$\alpha = -3.$$

$$\therefore P(x) = (x-2)^3(x+3)$$

①

MATHEMATICS Extension 2: Question...3...

Suggested Solutions

Marks

Marker's Comments

a) $\omega^3 - 1 = 0$

$(\omega - 1)(\omega^2 + \omega + 1) = 0$

ω is complex $\therefore \omega \neq 1$

$\therefore \omega^2 + \omega + 1 = 0$

$1 + \omega = -\omega^2$

$(1 + \omega - \omega^2)^7 = (-\omega^2 - \omega^2)^7$

$= (-2\omega^2)^7$

$= -128(\omega^2)^7$

$= -128 \omega^{14}$

$= -128 \omega^2 \times (\omega^3)^4$

$= -128 \omega^2$

1

b) $z^3 = -8$

let $z = r(\cos \theta + i \sin \theta)$

$\therefore z^3 = r^3 (\cos \theta + i \sin \theta)^3$

$= r^3 (\cos 3\theta + i \sin 3\theta)$ (De Moivre's)

$-8 = 8(\cos \pi + i \sin \pi)$

$\therefore r^3 (\cos 3\theta + i \sin 3\theta) = 8(\cos \pi + i \sin \pi)$

$\therefore r^3 = 8$

$r = 2$

$3\theta = \pi + 2k\pi$ where $k \in \mathbb{Z}$

$\theta = \frac{\pi + 2k\pi}{3}$

3

when $k=0$, $z_1 = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

$k=1$, $z_2 = 2(\cos \pi + i \sin \pi)$

$k=-1$, $z_3 = 2(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}))$

$\therefore$ roots are $2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$,

$2(\cos \pi + i \sin \pi)$, $2(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}))$

1

MATHEMATICS Extension 2: Question...3

Suggested Solutions

Marks

Marker's Comments

c) $\vec{AD} = -i \vec{AB}$ ($\vec{AD}$ is $\vec{AB}$ rotated clockwise by $\frac{\pi}{2}$)

$$d-a = -i(b-a)$$

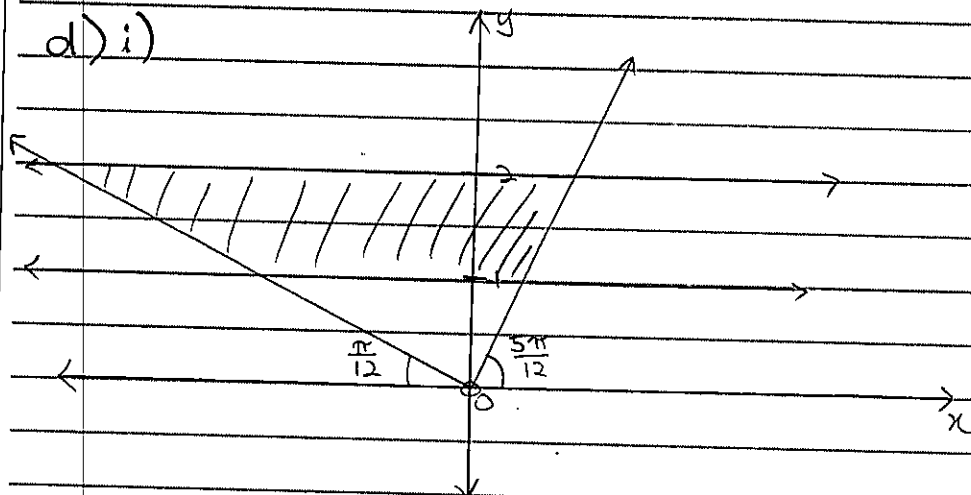
$$d = a - ib + ia$$

$$= a(1+i) - ib$$

1

1

d) i)



1 showing angles

1 correct region

1 correct lines

ii) $|z_0| = \frac{2}{\sin \frac{\pi}{12}}$

$$\arg(z_0) = \frac{11\pi}{12}$$

$$\therefore z_0 = \frac{2}{\sin \frac{\pi}{12}} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

$$= \frac{2}{\sin \frac{11\pi}{12}} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

$$\text{since } \sin \frac{\pi}{12} = \sin \frac{11\pi}{12}$$

$$= 2 \cot \frac{11\pi}{12} + 2i$$

$$\frac{2}{\tan \frac{11\pi}{12}} = \frac{2}{\tan(\frac{\pi}{4} + \frac{2\pi}{3})}$$

$$= \frac{2(1 - \tan \frac{\pi}{4} \tan \frac{2\pi}{3})}{\tan \frac{\pi}{4} + \tan \frac{2\pi}{3}}$$

$$= \frac{2(1 + \sqrt{3})}{(1 - \sqrt{3})}$$

$$= -4 - 2\sqrt{3}$$

1

1

exact form is:

$$z_0 = -4 - 2\sqrt{3} + 2i$$

MATHEMATICS Extension 2: Question...3...

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{e) i) } z^n &= (\cos \theta + i \sin \theta)^n \\ &= \cos n\theta + i \sin n\theta \text{ (De Moivre's) } \textcircled{1} \\ z^{-n} &= (\cos \theta + i \sin \theta)^{-n} \\ &= (\cos(-n\theta) + i \sin(-n\theta)) \\ &\quad \text{(De Moivre's)} \\ &= \cos n\theta - i \sin n\theta \textcircled{2} \end{aligned}$$

$$\text{since } \cos(-n\theta) = \cos n\theta.$$

$$\sin(-n\theta) = -\sin n\theta.$$

$$\therefore \textcircled{1} + \textcircled{2} :$$

$$\begin{aligned} z^n + z^{-n} &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta. \end{aligned}$$

$$\therefore \cos n\theta = \frac{z^n + z^{-n}}{2}$$

$$\textcircled{1} - \textcircled{2}$$

$$\begin{aligned} z^n - z^{-n} &= \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta \\ &= 2i \sin n\theta \end{aligned}$$

$$\therefore \sin n\theta = \frac{z^n - z^{-n}}{2i}$$

$$\text{ii) } 32 \sin^4 \theta \cos^2 \theta = \cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2$$

$$\text{LHS} = 32 \left(\frac{z - z^{-1}}{2i} \right)^4 \left(\frac{z + z^{-1}}{2} \right)^2$$

$$= 32 \frac{(z - z^{-1})^4}{16} \frac{(z + z^{-1})^2}{4}$$

$$= \frac{1}{2} (z - z^{-1})^2 (z + z^{-1})^2 (z - z^{-1})^2$$

$$= \frac{1}{2} (z^2 - z^{-2})^2 (z^2 - 2 + z^{-2})$$

$$= \frac{1}{2} (z^4 - 2 + z^{-4}) (z^2 - 2 + z^{-2})$$

$$= \frac{1}{2} (z^6 - 2z^4 + z^2 - 2z^2 + 4 - 2z^{-2} + z^{-2} - 2z^{-4} + z^{-6})$$

$$= \frac{1}{2} (z^6 + z^{-6} - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4)$$

$$= \frac{z^6 + z^{-6}}{2} - 2 \frac{(z^4 + z^{-4})}{2} - \frac{z^2 + z^{-2}}{2} + 2$$

using part (i)

expanding

1 $\Rightarrow$ changing back to correct form

MATHEMATICS Extension 2: Question...3....

Suggested Solutions

Marks

Marker's Comments

$$= \cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2$$

$$= \text{RHS}$$

OR

$$= \frac{1}{2} (z - z^{-1})^4 (z + z^{-1})^2$$

$$= \frac{1}{2} (z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4})$$

$$(z^2 + 2 + z^{-2})$$

$$= \frac{1}{2} (z^6 + 2z^4 + z^2 - 4z^4 - 8z^2 - 4$$

$$+ 6z^2 + 12 + 6z^{-2} - 4 - 8z^{-2}$$

$$- 4z^{-4} + z^{-2} + 2z^{-4} + z^{-6})$$

$$= \frac{1}{2} (z^6 + z^{-6} - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4)$$

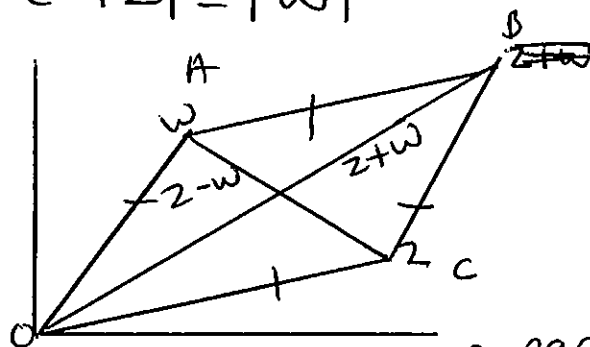
← different expansion
all else is
the same.

Suggested Solutions

Marks

Marker's Comments

Since $|z| = |w|$



OACB is a parallelogram

~~arg~~ since $|z| = |w|$

OACB is a rhombus
(4 equal sides)

Diagonals of a rhombus bisect
at right angles

$$\therefore \arg(z+w) - \arg(z-w) = \pm \frac{\pi}{2}$$

so $\frac{z+w}{z-w}$ is purely imaginary

1 for explaining
why a rhombus
1 for rhombus
diagonals
bisect at
right angles
(poorly done)

Students needed
to read whole
question to help
approach
question.

1 for ~~diagonal~~
arg statement
proving purely
imaginary.

* Method 2

$$\frac{(x+a) + i(y+b)}{(x-a) + i(y-b)} \times \frac{(x-a) - i(y-b)}{(x-a) - i(y-b)}$$

$$\frac{x^2 - a^2 + y^2 - b^2 + i(y+b)(x-a) - i(x+a)(y-b)}{(x-a)^2 + (y-b)^2}$$

Since $|z| = |w|$ $x^2 - a^2 + y^2 - b^2 = 0$
 $\therefore \operatorname{Re}(z) = 0$ so purely imaginary

1 for correct
start.

1 for using
 $|z| = |w|$

1 for geometry

Suggested Solutions

Marks

Marker's Comments

~~$$z = \sqrt{6} - \sqrt{2}i$$~~

$$z = \sqrt{6} - \sqrt{2}i$$

$$\arg(z) = \tan^{-1} \frac{-1}{\sqrt{3}}$$

$$|z| = \sqrt{6+2}$$

$$= 2\sqrt{2}$$

$$= -\frac{\pi}{6}$$

$$\therefore \sqrt{6} - \sqrt{2}i = 2\sqrt{2} \left(\cos\left(\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right) \right)$$

$$(1) \left(\frac{\sqrt{6} - \sqrt{2}i}{2 - 2i} \right)^{2000}$$

$$\left(\frac{2\sqrt{2} \cos\left(\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{6}\right)}{2\sqrt{2} \cos\left(-\frac{\pi}{6}\right) + i \sin\left(-\frac{\pi}{4}\right)} \right)^{2000}$$

$$= \left[\cos\left(-\frac{\pi}{6} + \frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{6} + \frac{\pi}{4}\right) \right]^{2000}$$

$$= \cos \frac{500\pi}{3} + i \sin \frac{500\pi}{3} \text{ (De Moivre)}$$

$$= \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)$$

$$= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

NB! A 2 mark question requires working

1 for correct

2 for correct answer

$2\sqrt{2} \text{cis}\left(-\frac{\pi}{6}\right)$
is not accepted in HSC. ~~1~~
lost a mark.

①

①

$$c) z^7 = 1$$

$$\text{Since } \alpha \text{ a root: } \alpha^7 = 1$$

$$\alpha^7 - 1 = 0$$

$$(\alpha - 1)(1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6) = 0$$

$$1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$$

(α complex)

$$(i) \quad \theta + f = \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6$$

$$= -1 \text{ (from part i)}$$

$$(\theta + f) = (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6)$$

$$= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8$$

$$+ \alpha^7 + \alpha^9 + \alpha^{10}$$

since $\alpha^7 = 1$

$$= 3 + \alpha^4 + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6$$

$$= 3 - 1$$

$$= 2$$

quadratic equation

$$x^2 + x + 2 = 0$$

well done.

Suggested Solutions

Marks

Marker's Comments

$$z^7 = 1$$

$$z^7 = \text{cis } 2k\pi$$

$$\therefore z = \text{cis } \frac{2k\pi}{7} \text{ (all mod } 7\text{)}$$

$$\text{Solving } x^2 + x + 2 = 0$$

$$x = \frac{-1 \pm \sqrt{7}}{2}$$

$$\theta = \alpha + \alpha^2 + \alpha^4$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} + \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}$$

Equating real

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2}$$

$$\text{but } \cos \frac{8\pi}{7} = \cos \left(\pi + \frac{\pi}{7} \right) = -\cos \frac{\pi}{7}$$

$$\therefore \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$$

Equating imaginary

$$\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$$

$$\text{Since } \sin \frac{2\pi}{7} > \sin \frac{\pi}{7} \text{ and } \sin \frac{2\pi}{7}, \sin \frac{4\pi}{7} > 0$$

Students who didn't find this struggled with second part of question

needed to explain why positive